Rerandomization simulation report

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Process for rerandomization simulation

To generate simulation data I did the following for each outcome j in our set of outcomes across our empirical dataset (excepting a few studies that were inaccessible):

1) Load the empirical data for outcome j, patch missing data, etc.
2) Calculate the empirical estimates of the treatment effects for outcome j, and the associated empirical standard errors.
3) Repeatedly, for simulation runs $r = 1, \ldots, R$, permute the treatment assignment across clusters and within blocks. I then re-estimated the impacts given the permuted assignment.
4) I then, within each run and outcome, calculated the range of estimates, and the ratio of the largest to smallest standard error, across all the estimators that were defined for that run. I did this both overall, and for the group of estimators within each estimand (Person or Cluster).
5) I finally saved the set of results to a file.

Due to the permutation, there is no actual treatment effect. Any actual treatment effect is pushed into individual and cluster variation. For example, if a cluster had a positive average effect, the permutation simulation views this cluster as just having a slightly higher intercept than it would have had, had it been a control cluster in reality. The structure of the data, the relationship of covariates to outcome, etc., is all preserved by this approach.

The datasets missing from this analysis

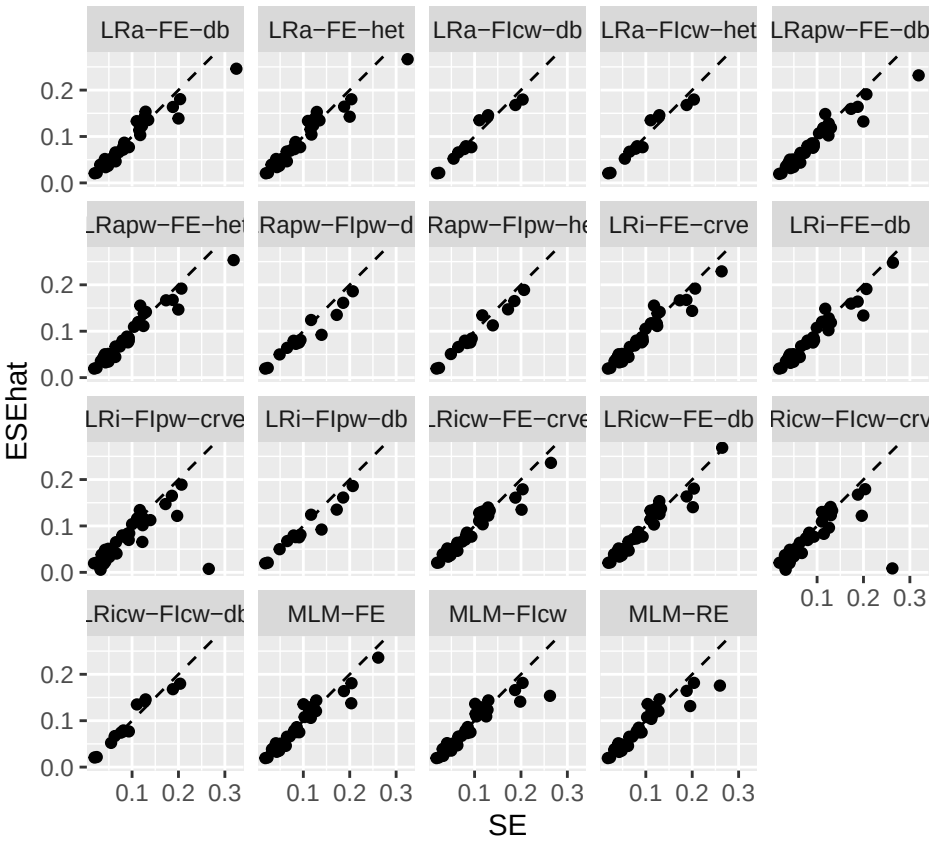
We didn't run the simulation on all the empirical datasets due to privacy restrictions and computational feasibility. The datasets dropped are the following:

```
##
##
## Table: Dropped Datasets
##
## |UniqueID |  n|  J|
## |:--------|--:|--:|
```

How do the simulation estimated SEs compare to the actual simulation SE?

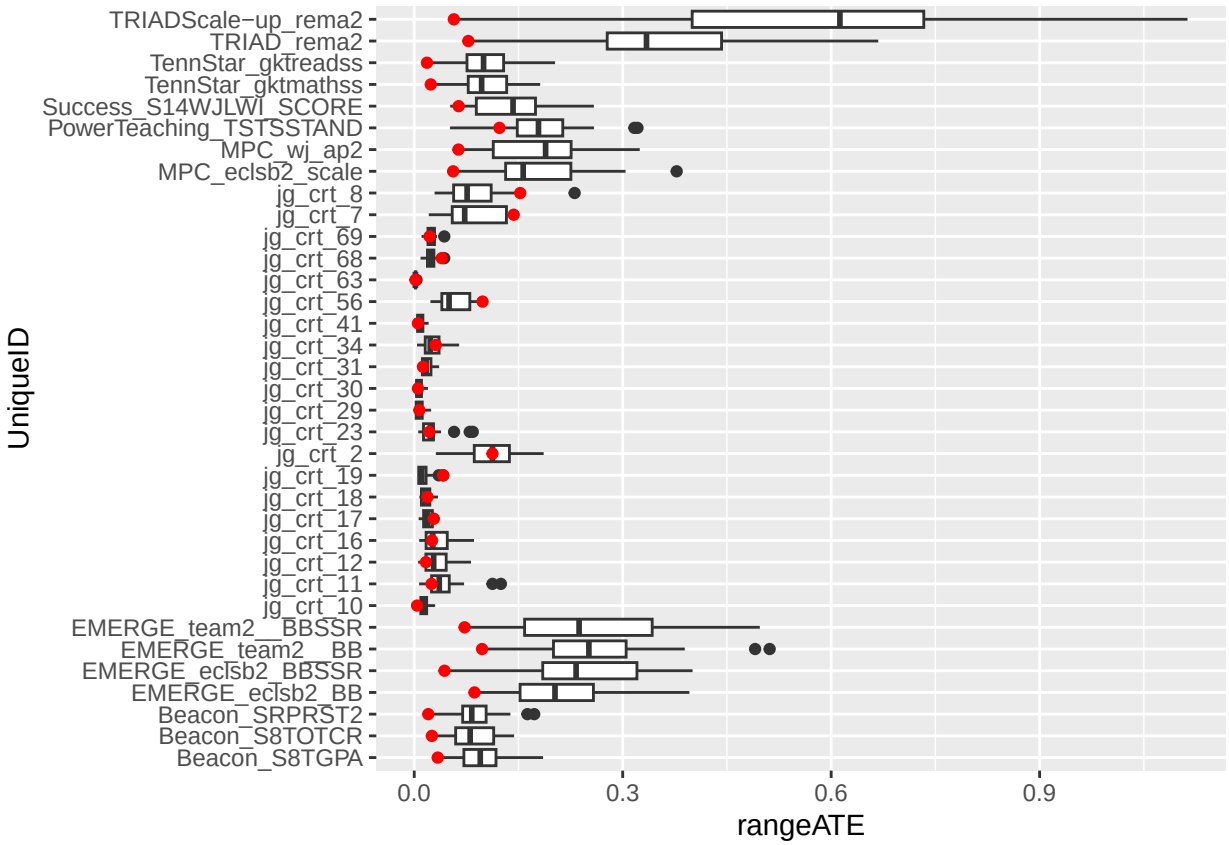
Within the simulation, we can compare the estimated SE to the standard deviation of estimates across the simulations (including the actual empirical data as a "simulation trial").

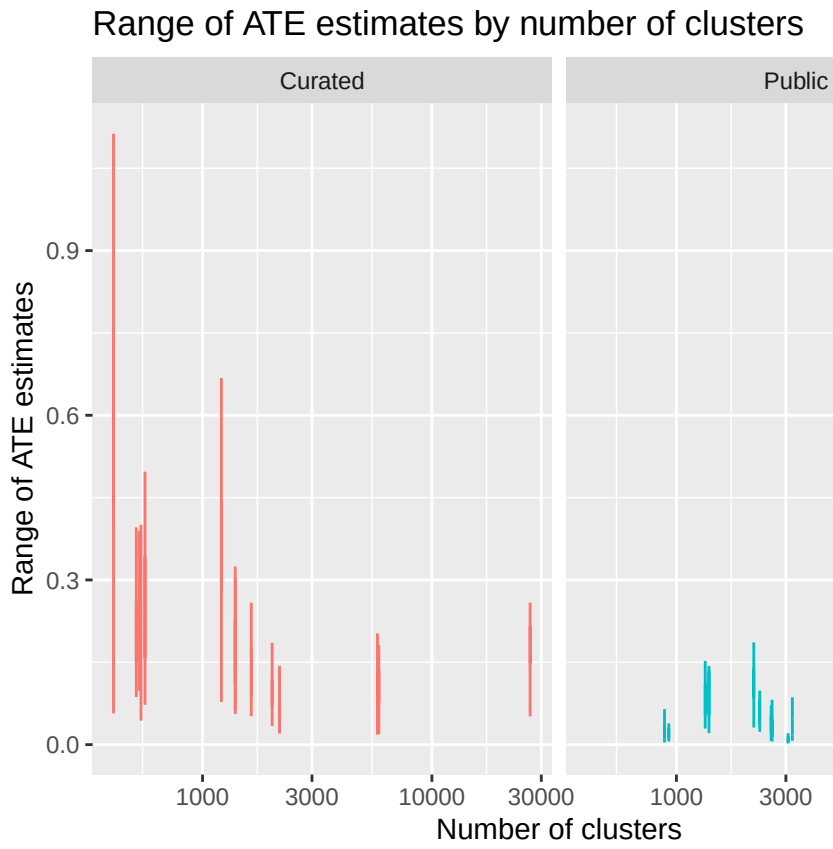
```
## Warning: Removed 132 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



The actual vs. permutation estimates

For each outcome, and across the simulation runs we can look at the actual ATE range vs. the ranges we get from the permutation. Weirdly, for the curated data, all the actual ranges are quite low. We also see notable differences between the two sets of data, with curated having much wider ranges in general.



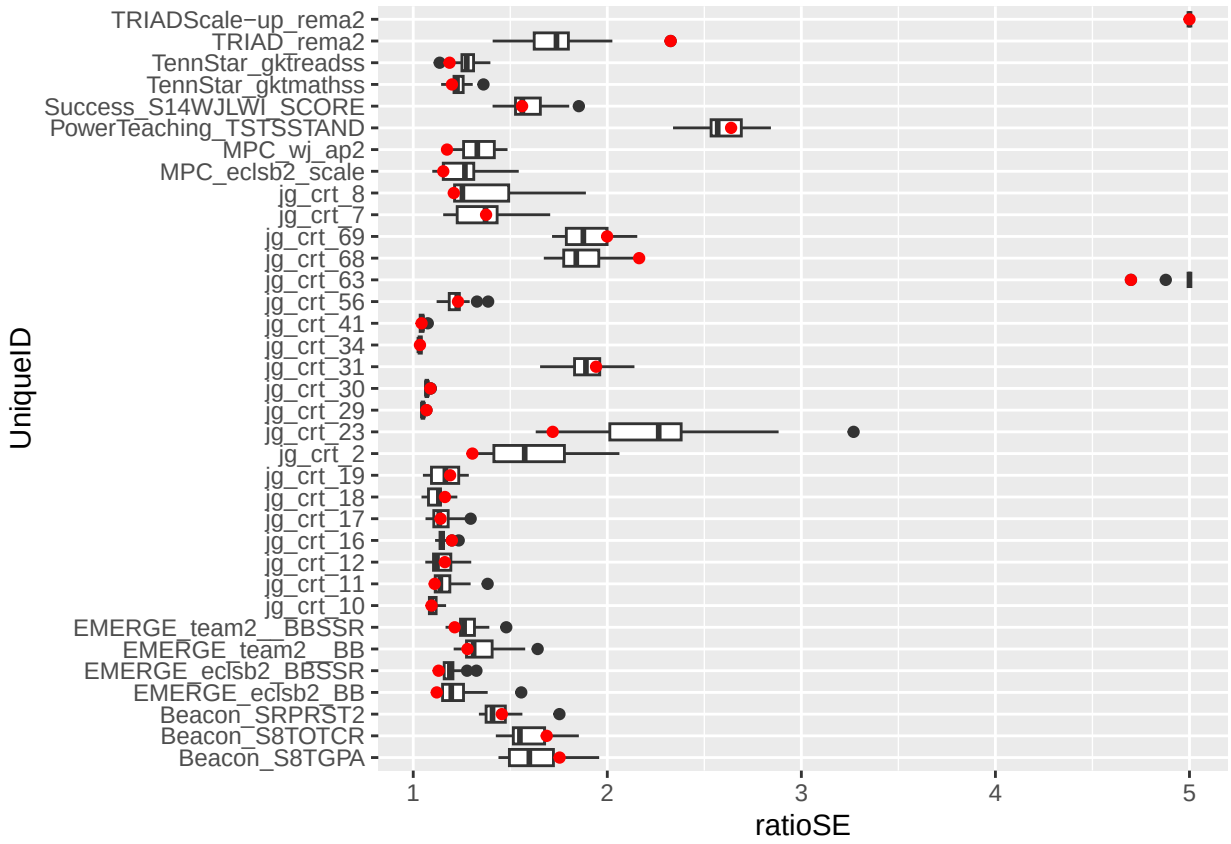


We can look at the studies by size (number of clusters):

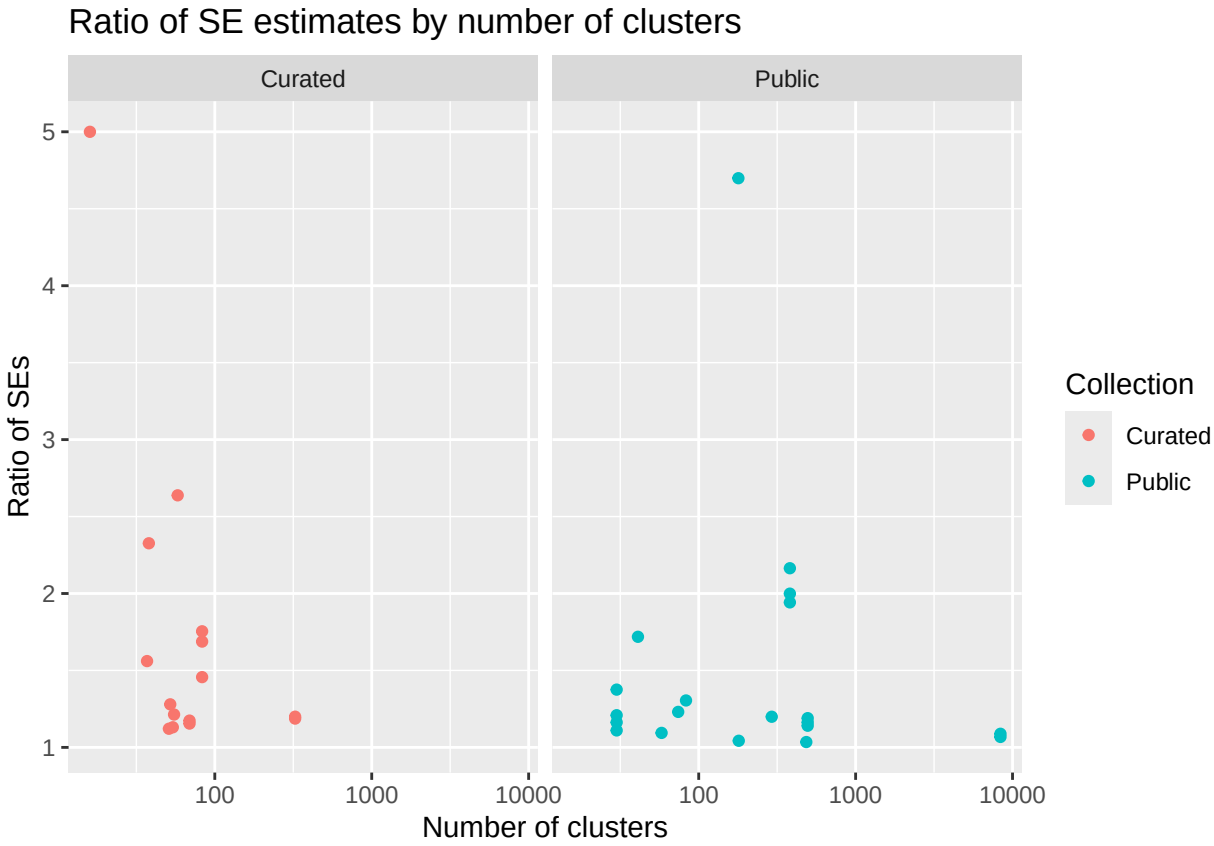


Variation in Standard Errors

We next look at how much the ratio of the max to min estimated standard errors vary across the permutations


TRIADScale−up_rema2
TRIAD_rema2
TennStar_gktreadss
TennStar_gktmathss
Success_S14WJLWI_SCORE
PowerTeaching_TSTSSTAND
MPC_wj_ap2
MPC_eclsb2_scale
jg_crt_8
jg_crt_7
jg_crt_69
jg_crt_68
jg_crt_63
jg_crt_56
jg_crt_41
jg_crt_34
jg_crt_31
jg_crt_30
jg_crt_29
jg_crt_23
jg_crt_2
jg_crt_19
jg_crt_18
jg_crt_17
jg_crt_16
jg_crt_12
jg_crt_11
jg_crt_10
EMERGE_team2__BBSSR
EMERGE_team2__BB
EMERGE_eclsb2_BBSSR
EMERGE_eclsb2_BB
Beacon_SRPRST2
Beacon_S8TOTCR
Beacon_S8TGPA
UniqueID
1
2
3
4
5
ratioSE


Ratio of SE estimates by number of clusters
Curated
Public
Range of ATE estimates
1
2
3
4
5
1000
3000
10000
30000
Number of clusters
Collection
Curated
Public



Imposter Plots

We can look at all the runs for all the datasets, and then for each run id r, make a histogram of the ranges across all the datasets, mimicking our core figures from the empirical paper. We then insert the real estimated values as an "imposter" so we could see if the estimated distributions looked like the actual one.

Aggregate statistics

For each run, we can look at the mean and standard deviation of the ranges of estimates across the datasets in that run.

Commentary